aiPHeed: food-insecurity risk indicators for CALABARZON (DA Region IV-A)

Model aiPHeed-E53-A1 (LightGBM: E48-A1 inputs + measured driver conditions, shrinkage calibration) - PREVIEW E53.A1+C2. Forecast quarter 2027-Q1, issued at origin 2026-10-01 with information published before it. Record SHA-256 5302cfb6219d5d94c1cab7401bfa61f9e00523a56e7df1a2ef7ae26a9f0219a6.
aiPHeed produces potential food-insecurity risk indicators for review. They are not official food-insecurity classifications and do not replace assessments by authorized government agencies.

What is estimated

T1: S1 = H + 2D > tau, HIGH or LOW (the approved Chapter I composite food-stress label)
H: SWS Balance Luzon self-rated hunger (%), one regional value for all five provinces (measured)
D: province food inflation minus the five-province mean (pp), PSA provincial CPI (measured)
Not: an official or measured food-insecurity prevalence; provincial differences come only from prices
Evidence status: EXPERIMENTAL / PREVIEW (E53: eligible under its pre-declared rule; no paired difference with E49 is significant)

CALABARZON: official regional hunger (measured)

Quarter	Round	Hunger %	Posted
2024-Q2	Second Quarter 2024 SWS Survey, June 23-July	19.6	2024-07-23
2024-Q3	September 2024 SWS Survey	18.1	2024-10-17
2024-Q4	December 2024 SWS Survey	25.3	2025-01-14
2025-Q1	Stratbase-SWS March 15-20, 2025 National Surv	24.0	2025-03-29
2025-Q4	November 2025 SWS Survey	16.7	2026-01-14
2026-Q1	March 2026 SWS Survey	22.4	2026-05-12
2026-Q2	June 2026 SWS Survey	17.3	2026-08-24
2026-Q3	September 2026 SWS Survey	23.6	2026-10-14

regional: ONE value shared by all five provinces; not a provincial or municipal measurement

Cavite: 2027-Q1 forecast (next quarter, issued 2026-10-01)

	2027-Q1 FORECAST	2026-Q3 backtest (observed HIGH)
P(HIGH)	82%	90%
Signal	STRONG	STRONG

Main drivers (TreeSHAP, log-odds):

Factor	Value	Contribution
Measured market pressure: rank of provincial food and rice inflation (	90.48	+0.62
Measured unemployment pressure (national, PSA LFS): rank against earli	75.07	+0.49
Province food-price pressure last quarter (PSA food CPI)	0.87	+0.45
Province food-price pressure in the first two months of this quarter	0.06	+0.41
Latest regional hunger relative to the HIGH threshold	2.31	+0.34
Relevant food-insecurity news in the region (count)	2.30	+0.11

exact TreeSHAP contribution of each input to this estimate, in log-odds (positive raises the probability of HIGH); associations the model learned from 2021-2026 data, not causes. Probabilities are calibrated by shrinkage toward the training base rate (E49-C2 procedure); driver shares come from the LightGBM before calibration. Employment and OFW Remittance inputs are national-level indicators.

The model puts Cavite at 82% for 2027-Q1. Raised mainly by: Measured market pressure: rank of provincial food and rice inflation (PSA CPI) against earlier quarters; Measured unemployment pressure (national, PSA LFS): rank against earlier quarters; Province food-price pressure last quarter (PSA food CPI).

Food inflation in Cavite was 0.47 pp above the five-province average in 2026-Q3 (PSA, measured).

News evidence (2026-Q3): 0 validated article(s) of 57 province-attributed; status NO_USABLE_EVIDENCE.

Review: the model estimate is high; check provincial market prices and field reports. Prioritise for review by DA Region IV-A; do not allocate resources on this estimate alone.

Top LGUs (relative position)	Risk index	Poverty used	Kind
Gen. Mariano Alvarez	0.5197	9.540674	PSA municipal SAE estimate
General Emilio Aguinaldo	0.4977	16.0907	PSA municipal SAE estimate
Cavite City	0.4925	12.708223	PSA municipal SAE estimate
Tagaytay	0.4743	15.283253	PSA municipal SAE estimate

Mendez	0.4574	15.108867	PSA municipal SAE estimate

Laguna: 2027-Q1 forecast (next quarter, issued 2026-10-01)

	2027-Q1 FORECAST	2026-Q3 backtest (observed HIGH)
P(HIGH)	82%	90%
Signal	STRONG	STRONG

Main drivers (TreeSHAP, log-odds):

Factor	Value	Contribution
Measured market pressure: rank of provincial food and rice inflation (	61.90	+0.62
Measured unemployment pressure (national, PSA LFS): rank against earli	75.07	+0.49
Province food-price pressure last quarter (PSA food CPI)	0.10	+0.45
Province food-price pressure in the first two months of this quarter	0.46	+0.41
Latest regional hunger relative to the HIGH threshold	2.31	+0.34
Relevant food-insecurity news in the region (count)	2.30	+0.11

exact TreeSHAP contribution of each input to this estimate, in log-odds (positive raises the probability of HIGH); associations the model learned from 2021-2026 data, not causes. Probabilities are calibrated by shrinkage toward the training base rate (E49-C2 procedure); driver shares come from the LightGBM before calibration. Employment and OFW Remittance inputs are national-level indicators.
The model puts Laguna at 82% for 2027-Q1. Raised mainly by: Measured market pressure: rank of provincial food and rice inflation (PSA CPI) against earlier quarters; Measured unemployment pressure (national, PSA LFS): rank against earlier quarters; Province food-price pressure last quarter (PSA food CPI).
Food inflation in Laguna was 0.27 pp above the five-province average in 2026-Q3 (PSA, measured).
News evidence (2026-Q3): 7 validated article(s) of 52 province-attributed; status SUFFICIENT.
Review: the model estimate is high; check provincial market prices and field reports. Prioritise for review by DA Region IV-A; do not allocate resources on this estimate alone.

Top LGUs (relative position)	Risk index	Poverty used	Kind
Siniloan	0.5043	24.356336	PSA municipal SAE estimate
San Pedro	0.383	5.987922	PSA municipal SAE estimate

Biñan	0.3775	5.750266	PSA municipal SAE estimate
Cabuyao	0.2214	5.015342	PSA municipal SAE estimate
Santa Cruz	0.2167	9.748186	PSA municipal SAE estimate

Batangas: 2027-Q1 forecast (next quarter, issued 2026-10-01)

	2027-Q1 FORECAST	2026-Q3 backtest (observed HIGH)
P(HIGH)	71%	89%
Signal	MODERATE	STRONG

Main drivers (TreeSHAP, log-odds):

Factor	Value	Contribution
Measured market pressure: rank of provincial food and rice inflation (	33.33	-0.96
Province food-price pressure last quarter (PSA food CPI)	-2.43	-0.59
Province food-price pressure in the first two months of this quarter	-2.19	-0.58
Measured unemployment pressure (national, PSA LFS): rank against earli	75.07	+0.49
Latest regional hunger relative to the HIGH threshold	2.31	+0.34
Relevant food-insecurity news in the region (count)	2.30	+0.11

exact TreeSHAP contribution of each input to this estimate, in log-odds (positive raises the probability of HIGH); associations the model learned from 2021-2026 data, not causes. Probabilities are calibrated by shrinkage toward the training base rate (E49-C2 procedure); driver shares come from the LightGBM before calibration. Employment and OFW Remittance inputs are national-level indicators.
The model puts Batangas at 71% for 2027-Q1. Lowered mainly by: Measured market pressure: rank of provincial food and rice inflation (PSA CPI) against earlier quarters; Province food-price pressure last quarter (PSA food CPI); Province food-price pressure in the first two months of this quarter.
Food inflation in Batangas was 2.63 pp below the five-province average in 2026-Q3 (PSA, measured).
News evidence (2026-Q3): 0 validated article(s) of 73 province-attributed; status NO_USABLE_EVIDENCE.
Monitor: the estimate is above one half with less certainty. Prioritise for review by DA Region IV-A; do not allocate resources on this estimate alone.

Top LGUs (relative position)	Risk index	Poverty used	Kind
Lipa	0.6393	12.027362	PSA municipal SAE estimate
Bauan	0.5318	10.654363	PSA municipal SAE estimate

San Pascual	0.5298	11.372837	PSA municipal SAE estimate
Taal	0.4881	9.453046	PSA municipal SAE estimate
San Jose	0.4445	9.862561	PSA municipal SAE estimate

Rizal: 2027-Q1 forecast (next quarter, issued 2026-10-01)

	2027-Q1 FORECAST	2026-Q3 backtest (observed HIGH)
P(HIGH)	82%	90%
Signal	STRONG	STRONG

Main drivers (TreeSHAP, log-odds):

Factor	Value	Contribution
Measured market pressure: rank of provincial food and rice inflation (	66.67	+0.62
Measured unemployment pressure (national, PSA LFS): rank against earli	75.07	+0.49
Province food-price pressure last quarter (PSA food CPI)	0.47	+0.45
Province food-price pressure in the first two months of this quarter	0.51	+0.41
Latest regional hunger relative to the HIGH threshold	2.31	+0.34
Relevant food-insecurity news in the region (count)	2.30	+0.11

exact TreeSHAP contribution of each input to this estimate, in log-odds (positive raises the probability of HIGH); associations the model learned from 2021-2026 data, not causes. Probabilities are calibrated by shrinkage toward the training base rate (E49-C2 procedure); driver shares come from the LightGBM before calibration. Employment and OFW Remittance inputs are national-level indicators.
The model puts Rizal at 82% for 2027-Q1. Raised mainly by: Measured market pressure: rank of provincial food and rice inflation (PSA CPI) against earlier quarters; Measured unemployment pressure (national, PSA LFS): rank against earlier quarters; Province food-price pressure last quarter (PSA food CPI).
Food inflation in Rizal was 0.71 pp above the five-province average in 2026-Q3 (PSA, measured).
News evidence (2026-Q3): 2 validated article(s) of 40 province-attributed; status LIMITED_SIGNAL.
Review: the model estimate is high; check provincial market prices and field reports. Prioritise for review by DA Region IV-A; do not allocate resources on this estimate alone.

Top LGUs (relative position)	Risk index	Poverty used	Kind
Taytay	0.82	9.808941	PSA municipal SAE estimate

Antipolo	0.539	9.294249	PSA municipal SAE estimate
Cardona	0.5246	9.233761	PSA municipal SAE estimate
Morong	0.4846	8.997131	PSA municipal SAE estimate
San Mateo	0.4505	6.644608	PSA municipal SAE estimate

Quezon: 2027-Q1 forecast (next quarter, issued 2026-10-01)

	2027-Q1 FORECAST	2026-Q3 backtest (observed HIGH)
P(HIGH)	82%	89%
Signal	STRONG	STRONG

Main drivers (TreeSHAP, log-odds):

Factor	Value	Contribution
Measured market pressure: rank of provincial food and rice inflation (	76.19	+0.62
Measured unemployment pressure (national, PSA LFS): rank against earli	75.07	+0.49
Province food-price pressure last quarter (PSA food CPI)	1.00	+0.45
Province food-price pressure in the first two months of this quarter	1.16	+0.41
Latest regional hunger relative to the HIGH threshold	2.31	+0.34
Relevant food-insecurity news in the region (count)	2.30	+0.11

exact TreeSHAP contribution of each input to this estimate, in log-odds (positive raises the probability of HIGH); associations the model learned from 2021-2026 data, not causes. Probabilities are calibrated by shrinkage toward the training base rate (E49-C2 procedure); driver shares come from the LightGBM before calibration. Employment and OFW Remittance inputs are national-level indicators.

The model puts Quezon at 82% for 2027-Q1. Raised mainly by: Measured market pressure: rank of provincial food and rice inflation (PSA CPI) against earlier quarters; Measured unemployment pressure (national, PSA LFS): rank against earlier quarters; Province food-price pressure last quarter (PSA food CPI).

Food inflation in Quezon was 1.17 pp above the five-province average in 2026-Q3 (PSA, measured).

News evidence (2026-Q3): 0 validated article(s) of 40 province-attributed; status NO_USABLE_EVIDENCE.

Review: the model estimate is high; check provincial market prices and field reports. Prioritise for review by DA Region IV-A; do not allocate resources on this estimate alone.

Top LGUs (relative position)	Risk index	Poverty used	Kind
Lucena	0.5657	16.8	province-mean substitute (no munic
San Francisco	0.5075	30.214238	PSA municipal SAE estimate
Macalelon	0.5048	30.101027	PSA municipal SAE estimate
San Andres	0.4631	27.817577	PSA municipal SAE estimate
Infanta	0.4543	27.335361	PSA municipal SAE estimate

Model evaluation: one quarter ahead, development quarters

Nested walk-forward, lead 1, 13 development quarters; run E53-A1 + C2 calibrated backtests

Approach	Correct/n	Acc	Acc 95% CI	Bal.acc	Prec	Recall	F1	ROC-AUC	Brier
Proposed: combined LightGBM (government + news), nes	51/65	0.785	0.646-0.908	0.836	0.976	0.755	0.851	0.880	0.150
B1 Naive persistence	57/65	0.877	0.800-0.938	0.860	0.959	0.887	0.922	0.860	0.123
B2 Logistic regression, combined features	56/65	0.862	0.785-0.938	0.818	0.940	0.887	0.913	0.840	0.120
B3 Logistic regression, government indicators only	55/65	0.846	0.754-0.923	0.809	0.939	0.868	0.902	0.866	0.102
B4 Logistic regression, NLP features only	46/65	0.708	0.554-0.846	0.692	0.905	0.717	0.800	0.658	0.198
B5 LightGBM, government indicators only	55/65	0.846	0.769-0.923	0.841	0.957	0.849	0.900	0.870	0.145
B6 LightGBM, NLP features only	44/65	0.677	0.492-0.846	0.673	0.900	0.679	0.774	0.736	0.195
Diagnostic: always HIGH (majority class)	53/65	0.815	0.677-0.923	0.500	0.815	1.000	0.898	0.500	0.185

One-quarter-ahead accuracy of the proposed LightGBM on 65 development estimates (13 quarters, 2023-Q1 to 2026-Q3): 0.785. The 95% date-bootstrap interval (0.646 to 0.908) reaches below 75%, so the target is not demonstrated. Always-HIGH reaches 0.815 and persistence 0.877. Unlike always-HIGH, the model identifies LOW quarters (balanced accuracy 0.836). No paired difference with any of the six baselines is significant. These are development dates; the 2026-Q4 forecast is the first untouched test.

Appendix: lead-2 experiment (not served)

Nested walk-forward, lead 2, 11 development quarters (the 55 rows of E38); run 20261016T122641Z

Approach	Correct/n	Acc	Acc 95% CI	Bal.acc	LOW recall	Brier
Diagnostic: always HIGH (majority class)	49/55	0.891	0.818-0.964	0.500	0/6	0.109
B1 Naive persistence	44/55	0.800	0.691-0.891	0.742	4/6	0.200
Previous lead-2 LightGBM (E38)	39/55	0.709	0.582-0.836	0.544	2/6	0.200
Proposed preview: E53-A1 at lead 2, shrinkage-calibr	37/55	0.673	0.545-0.800	0.670	4/6	0.214

Next-quarter accuracy 0.673 (37/55), balanced accuracy 0.670: below the 75% target and below persistence (0.800). It did not meet its adoption rule (E55). Shown as an experimental preview only.

News relevance validation (blind holdout)

Rule R1: precision 0.852, recall 0.395, F1 0.54.
Data sufficiency (news, FSSI-v2): {"collected": 8320, "province_attributed": 2820, "relevant_R1": 140, "syndicated_copies_removed": 0, "admitted_to_FSSI_v2": 140, "province_quarters_sufficient_v2": 7, "province_quarters_limited_v2": 46, "province_quarters_none_v2": 67, "in_FSSI_v1": 55, "province_quarters_sufficient_v1": 0}

City and municipal ranking

Indicative, relative ordering of the 142 cities and municipalities for planning attention. It is the province estimate re-weighted by PSA 2021 poverty incidence (60%) and 2024 POPCEN population density (40%) inside each province. It is not a measured municipal food-insecurity rate, not an independent municipal forecast and not an official classification (Chapter I, 1.7). The province estimate is experimental (not validated against baselines), so the ranking is EXPERIMENTAL.

- It does not say that any LGU is food insecure, or how many households are.
- Positions inside a province reflect the vulnerability weights only. Positions across provinces are not a measured comparison: the vulnerability index is scaled 0 to 1 inside each province, so the region-wide order mixes five different scales and follows the within-province index more than the province estimate. Compare LGUs within their province.
- For an LGU without a PSA municipal poverty estimate (Lucena City, a highly urbanized city that the PSA no-HUC table excludes), the poverty term is a province substitute, so its position inside the province comes from population density and the substitute value, not from a municipal measurement.
- Two LGUs with equal risk_index share a rank; their listing order (PSGC) carries no meaning.

Limitations

- The label is dominated by one regional SWS hunger value, so in most quarters all five provinces share a label; 53 of 65 development labels are HIGH. Provincial differences come mainly from relative food prices.
- One-quarter-ahead accuracy is 0.785 on 65 development estimates, below always-HIGH (0.815) and below persistence (0.877). The 95% interval reaches 0.646, so the 75% target is not demonstrated. No baseline is beaten significantly.
- The measured Employment and OFW Remittance inputs are national-level indicators (PSA LFS national unemployment; BSP national remittances, 2026 values provisional): they are the same for every province and cannot by themselves separate provinces.
- News signals added no measurable skill: the government-only LightGBM (B5) scores as well or better than the combined model on the same rows.
- News evidence is title-only (Google News RSS), thin in older quarters, and its relevance labels were made by the AI assistant; a researcher re-review is pending.
- LGU rankings re-weight the province forecast by PSA 2021 municipal poverty (141 of 142 LGUs; Lucena City, a highly urbanized city absent from the PSA no-HUC table, uses a province substitute of unverified source) and 2024 POPCEN density. They are relative positions, not municipal forecasts.
- The next-quarter forecast (two quarters after the latest complete data quarter; E55, E53-A1 at lead 2) scored 0.673 (37/55) on development dates, balanced accuracy 0.670, below the 75% target and below persistence (0.800); it did not meet its adoption rule and is shown as an experimental preview only.

- The 2026-Q4 and 2027-Q1 forecasts are the first untouched tests; they can be scored only after the SWS December 2026 and March 2027 rounds are released.

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